

Table 5: Hyperparameter Settings for RL Training Stage.

	Qwen2.5-VL-7B-Instruct-RL	InternVL2.5-38B-RL
pretrain model	Qwen2.5-VL-7B-Instruct	InternVL2.5-38B
RL Algorithm	rloo	rloo
train batch size	128	64
rollout batch_size	256	128
temperature	1	1
n samples per prompt	16	8
prompt max len	1024	4096
generate max len	3000	3000
bf16	True	True
actor learning rate	1e-6	1e-6
init kl coef	0	0

Other Details. The training environment consists of CentOS Linux release 7.6.1810 operating system with CUDA 12.1. For Qwen2.5-VL-7B-Instruct-RL, we train for 80 steps on 1×8 A800 GPUs and for InternVL2.5-38B-RL we train for 100 steps on 6×8 A800 GPUs.

C.5 RL models Evaluation Details

As mentioned above, we apply format rewards in RL experiments. Thus, to fully investigate the models’ latent reasoning abilities, we utilize implement training-aligned prompts during evaluation in VisuLogic, which is shown as follows: *“Solve the complex visual logical reasoning problem through step-by-step reasoning. Think about the reasoning process first and answer the question following this format: <think> THINKING </think><answer> ANSWER </answer>”*.

C.6 Effectiveness of RL Experiments

Figures 20, 21, 22, 23, 24 and 25 demonstrate qualitative differences in model outputs between baseline and RL optimized models. It illustrates reinforcement learning (RL) training enables the model to perform fundamental-level analysis of reasoning tasks embedded in graphical representations.

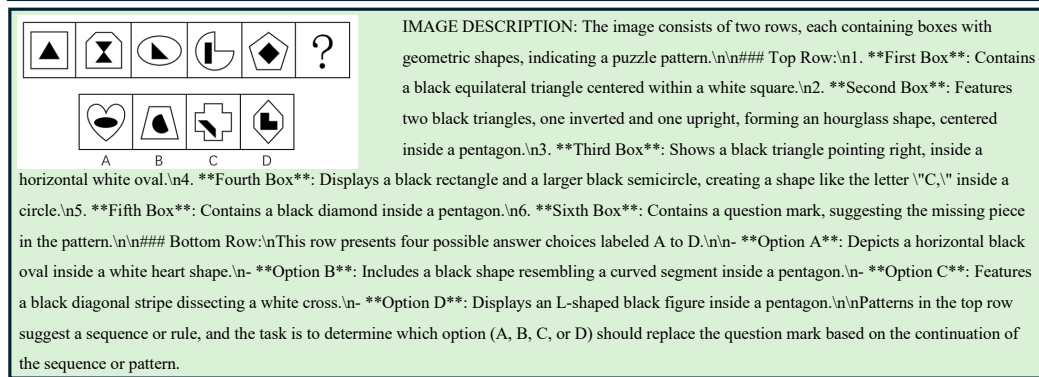
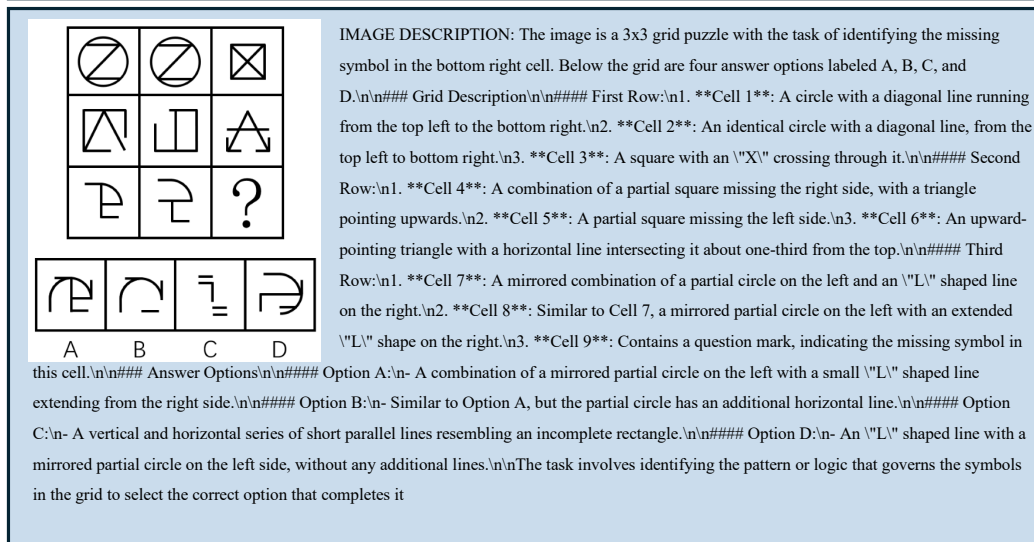
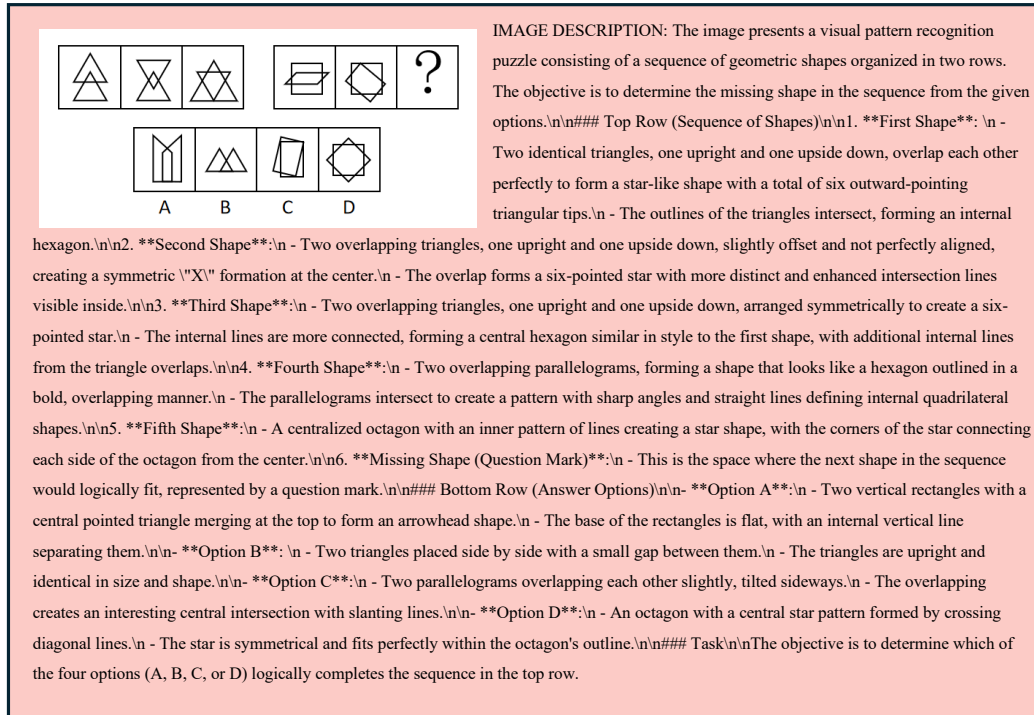


Figure 14: Part of image caption in LLM evaluation.

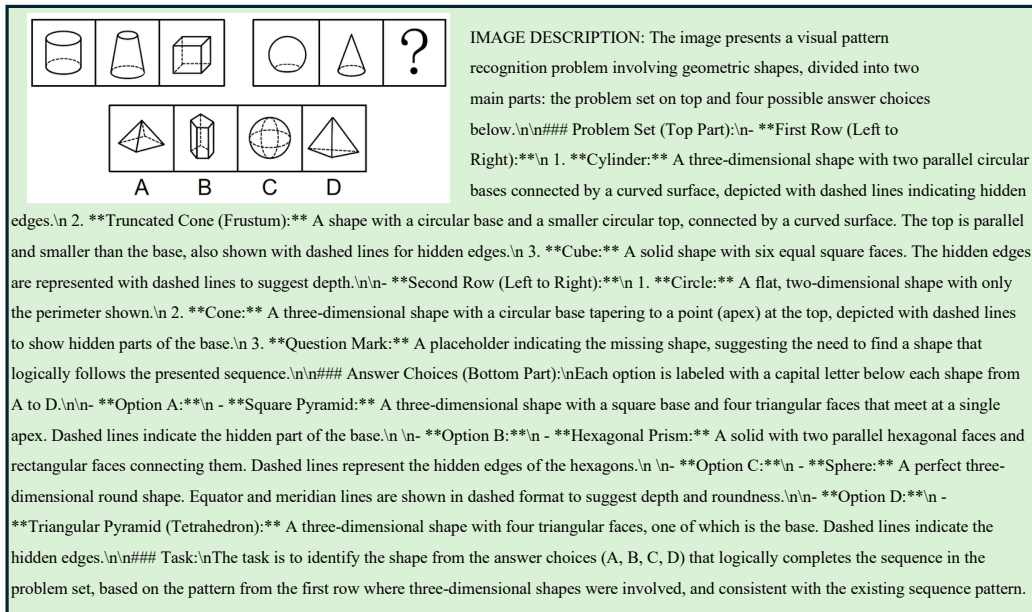
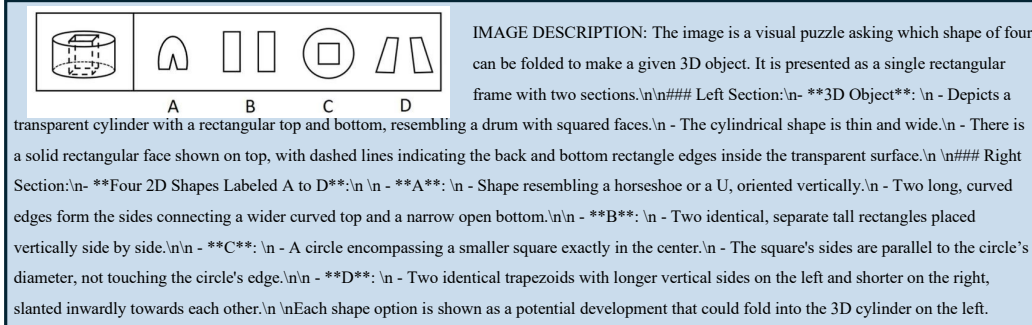
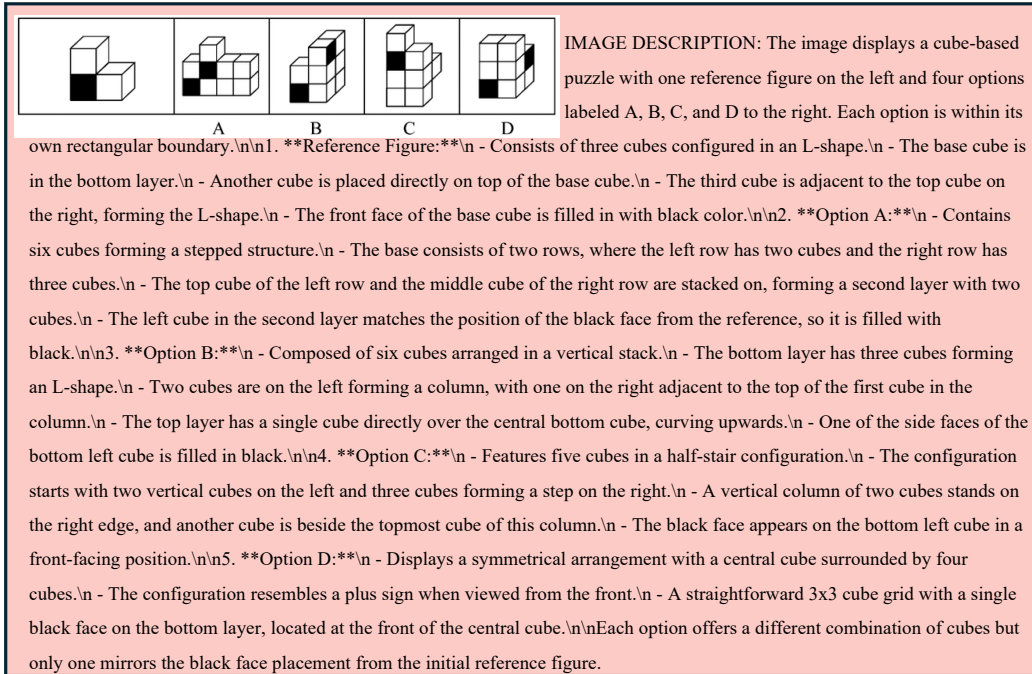


Figure 15: Part of image caption in LLM evaluation.